

FROST-GKR: Degree-7 Sumchecks via Frobenius Reduction and Shifted Tables for Poseidon2b over Binary Tower Fields

Breaking the Degree-2 Bottleneck in Sequential Hashing Arguments

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Abstract

Goldwasser-Kalai-Rothblum (GKR) protocols over binary fields have traditionally relied on degree-2 product sumchecks to prove non-linear operations, resulting in proof sizes and prover times that scale linearly with the circuit’s multiplicative depth. When applied to modern arithmetization-friendly hash functions like Poseidon2b, the decomposition of the degree-7 S-box ($x \mapsto x^7$) into degree-2 layers introduces severe overhead, both in round complexity and Fiat-Shamir transcript absorption.

In this paper, we present FROST-GKR (Frobenius Reduction Over Shifted Tables), a novel GKR construction that directly proves degree-7 identities over multilinear extensions (MLEs). FROST-GKR leverages two key techniques. First, it exploits the Frobenius endomorphism in $GF(2^{128})$ —where squaring is a linear operation—to evaluate the x^7 S-box constraint with only three multiplications, entirely eliminating the need for intermediate degree-2 decomposition. Second, it introduces the Shift Gadget, a sub-protocol utilizing a Change of Variable (CoV) to efficiently handle sequential linear constraints (such as MDS matrices and round constants) over the boolean hypercube.

Combined with a flat-basis polynomial representation for hardware acceleration, FROST-GKR collapses 59 Poseidon2b permutations—previously requiring 472 separate sumchecks—into a single unified sumcheck. In our internal implementation, dubbed the “Kill-Shot” protocol, FROST-GKR reduces the number of Fiat-Shamir rounds from 4,248 to just 30, fundamentally shifting the prover bottleneck from transcript hashing to ALU-bound arithmetic. We demonstrate a $10\times$ reduction in prover time, a $15\times$ reduction in verifier time, and a $50\times$ reduction in proof size compared to the legacy degree-2 baseline, establishing a new paradigm for arithmetizing high-degree operations over binary tower fields.

1 Introduction

The advent of transparent validity proofs (STARKs) over small fields has revitalized interest in the GKR protocol as a means to prove repetitive, structured computations efficiently. Systems like Binius have demonstrated the efficacy of binary tower fields $GF(2^{128})$ for polynomial IOPs due to their natural alignment with computer architectures and post-quantum security.

However, a fundamental bottleneck arises when proving computation traces involving arithmetization-oriented hash functions, specifically Poseidon2b. The Poseidon2b permutation consists of 66 rounds (8 external, 58 internal), each applying an S-box $x \mapsto x^7$ and a linear MDS layer. The standard GKR approach decomposes x^7 into a sequence of degree-2 constraints (e.g., $x^2 = x \cdot x$, $x^4 = x^2 \cdot x^2$, $x^3 = x^2 \cdot x$, $s_{out} = x^4 \cdot x^3$). For a 59-permutation spine, this decomposition yields 472 separate sumchecks. The sheer volume of Fiat-Shamir rounds (4,248) dominates the prover time, and the linear scaling of proof sizes inflates the transcript to hundreds of kilobytes.

We propose departing from the degree-2 dogma. We present a unified protocol that proves the degree-7 S-box constraint directly, and introduce a mathematical technique—the Shift Gadget—to cleanly interface sequential linear constraints with the high-degree sumcheck without sacrificing multilinearity.

1.1 Our Contributions

1. **Degree-7 Sumcheck via Frobenius:** We observe that in $GF(2^{128})$, the Frobenius endomorphism $(a + \alpha b)^2 = a^2 + \alpha^2 b^2$ allows computing x^7 with only 3 multiplications and 2 free squarings. This makes a degree-7 round polynomial significantly cheaper to evaluate than naively assumed, completely eliminating the need for intermediate degree-2 decomposition.
2. **The Shift Gadget:** Proving sequential dependencies ($s_{in}^{r+1} = MDS \cdot \pi^r + RC^{r+1}$) over a boolean hypercube is non-trivial, as index decrement ($dec(y)$) involves carry logic, making it a polynomial of degree 7 over the bits of y . We solve this by executing the main sumcheck over a shifted variable $y = inc(x)$, materializing shifted tables (e.g., *state_inc*), and delegating the consistency proof to a dedicated Shift Gadget sumcheck.
3. **Flat-Basis Accumulation:** We detail an implementation strategy where the 23 witness and public MLE tables are kept in the polynomial (flat) basis throughout the sumcheck fold, utilizing the PCLMULQDQ instruction for $O(1)$ field multiplications, and converting back to the tower basis only for transcript absorption.

2 Preliminaries

2.1 Binary Tower Fields

Let $F = GF(2^{128})$ constructed as a tower of extensions over $GF(2)$. An element $x \in F$ can be represented in the tower basis or the polynomial (flat) basis. The map $to_flat : F_{tower} \rightarrow F_{flat}$ is a linear isomorphism.

Lemma 1 (Frobenius Endomorphism). In $GF(2^{128})$, squaring is an F_2 -linear operation. For any $a, b \in F$ and scalar α , $(a + \alpha b)^2 = a^2 + \alpha^2 b^2$. Consequently, computing $x^7 = x \cdot x^2 \cdot x^4$ requires exactly 3 multiplications and 2 linear squarings.

2.2 Multilinear Extensions (MLE) and Sumcheck

Given a function $f : \{0, 1\}^n \rightarrow F$, its multilinear extension $\tilde{f} : F^n \rightarrow F$ is the unique polynomial of degree 1 in each variable agreeing with f on the boolean hypercube. The sumcheck protocol allows a prover to convince a verifier of the claim $\sum_{x \in \{0, 1\}^n} g(x) = v$ for a polynomial g . If g has maximum degree d per variable, the round polynomial requires $d + 2$ evaluation points.

2.3 Poseidon2b Permutation

We consider the Poseidon2b permutation over F^4 . It consists of $R = 66$ rounds. Each round applies an affine layer (Addition of Round Constants followed by an MDS matrix multiplication) and a non-linear S-box layer. In external rounds, the S-box is applied to all 4 state elements; in internal rounds, it is applied only to the first element.

3 The FROST-GKR Construction

We consider the proof of a sequence of Poseidon2b permutations (a “spine”) over a unified MLE.

3.1 MLE Layout and Identities

We define a 15-var MLE over the index $x = \text{slot} : 6 \parallel \text{round} : 7 \parallel \text{elem} : 2$. We maintain four witness MLEs: $s_{in}(x)$, $s_{out}(x)$, $state(x)$, and a public selector $\sigma(x)$ (1 where the S-box is active, 0 for identity bypass in internal rounds).

The identities to be proven per round x are:

1. **S-box** (C_1): $\sigma(x) \cdot (s_{out}(x) - s_{in}(x)^7) + (1 + \sigma(x)) \cdot (s_{out}(x) - s_{in}(x)) = 0$
2. **Round Constant** (C'_1): $\sigma(x) \cdot (s_{in}(x) - state(x) - RC(x)) = 0$
3. **MDS Transition** (C_2): $state(y) + \sum M_{kind(x)[e][j]} \cdot \pi(x|_{e=j}) = 0$ where $y = inc(x)$ (the next round index) and $\pi = \sigma \cdot s_{out} + (1 + \sigma) \cdot state$ handles the bypass of non-active elements in internal rounds.

3.2 The Change of Variable (CoV) and Shifted Tables

Constraint C_2 evaluates witness tables at $y = inc(x)$. Proving this directly over x requires evaluating MLEs at $inc(x)$, which involves carry logic and is a polynomial of degree 7 over the bits of x .

To avoid evaluating MLEs at this non-linear point in the hot loop, we perform a Change of Variable. We run the sumcheck over y and define the public weight function:

$$U(y) = eq(\rho, dec(y)) \cdot \mu(dec(y))$$

where μ is the “valid round” mask. We materialize shifted tables (e.g., $state_inc[y] = state[dec(y)]$, and similarly for s_{in_dec} , s_{out_dec} , σ_{dec} , etc.).

The main sumcheck proves:

$$\sum_y U(y) \cdot [C_1(dec(y)) + \beta C'_1(dec(y)) + \gamma C_2(y)] = 0$$

All constraints are evaluated using the shifted tables. The degree over y is capped at 7 (dictated solely by $s_{in_dec}^7$), resulting in a round polynomial of degree 9.

3.3 The Shift Gadget

The main sumcheck yields claims on the shifted tables (e.g., $\widetilde{state_inc}(r')$). We must prove consistency with the original committed MLEs (e.g., $state$).

We prove: $\widetilde{state_inc}(r') = \sum_x eq(r', inc(x)) \cdot \widetilde{state}(x)$. Since $inc(x)$ is degree 7 in x , the polynomial $eq(r', inc(x))$ has degree 7. Multiplied by the multilinear $\widetilde{state}(x)$, the Shift Gadget requires a sumcheck of degree 8 over 15 rounds.

While not a degree-1 sumcheck, this gadget operates on a single MLE ($\widetilde{state}$) rather than the 23 tables required by the main sumcheck, making it significantly cheaper. It reduces the shifted claim to a single point opening $\widetilde{state}(r'')$, which is batched with the other witness openings.

4 Implementation and Optimization

4.1 Flat-Basis Accumulation

Field multiplication in the tower basis requires recursive Karatsuba ($\sim 80\text{--}100$ ns). By mapping the 23 MLE tables to the flat basis via a linear map to_flat , we utilize the PCLMULQDQ instruction for carry-less multiplication with GCM reduction, reducing mul latency to $\sim 5\text{--}8$ ns.

The prover converts all tables to the flat basis prior to the sumcheck loop. The fold operation ($v_{lo} + t \cdot v_{hi}$) is basis-agnostic (linear). Only the final point claims are mapped back to the tower basis via to_tower for Fiat-Shamir absorption.

4.2 Polynomial Coefficient Accumulation

Instead of evaluating the degree-9 integrand at 10 separate points $\alpha \in \{0..9\}$ per cell, we express the integrand as a polynomial in α and accumulate its 10 monomial coefficients directly via XOR. This collapses the inner 10-iteration loop into a single cache-friendly pass over the MLE tables per round.

5 Performance Evaluation

We benchmark FROST-GKR for a 59-permutation Poseidon2b spine over $GF(2^{128})$ on a modern x86 CPU.

Metric	Legacy GKR (Degree-2)	FROST-GKR (Ours)	Improvement
Prover Time	1.63 s	154 ms	$10.5\times$
Verifier Time	1.06 s	69 ms	$15.3\times$
Proof Size	280 KB	5.44 KB	$50\times$
FS Rounds	4,248	30	$141\times$

Table 1: Performance comparison of FROST-GKR vs. Legacy degree-2 GKR.

Note: Verifier time is currently dominated by FRI opening verification. Transitioning to a Sumcheck-based PCS (Basefold) is projected to reduce verifier time to < 2 ms.

6 Security Analysis

6.1 Soundness

The soundness of FROST-GKR relies on the standard Schwartz-Zippel lemma. The prover commits to a set of multilinear polynomials. If the prover attempts to cheat on the unified sumcheck, the random challenges β and γ (drawn from the Fiat-Shamir transcript) act as a Random Linear Combination (RLC). By Schwartz-Zippel, the probability that a false constraint evaluates to zero at a random point is bounded by $d/|F|$, where $d \leq 9$ for the unified round polynomial. The combined soundness error for the unified sumcheck, shift gadget, and batch evaluation is bounded by $345/2^{128}$ (approximately 120 bits of security). The Shift Gadget similarly reduces to a sumcheck with negligible soundness error.

6.2 Fiat-Shamir Security

The legacy degree-2 GKR required 4,248 Fiat-Shamir rounds to prove the 59-permutation spine. While heuristically secure, an excessive number of rounds expands the attack surface for transcript grinding and hash-collision attacks. FROST-GKR reduces this to exactly 30 rounds (15 for the main sumcheck, 15 for the Shift Gadget), significantly tightening the Fiat-Shamir security bound and reducing the transcript size by orders of magnitude.

7 Related Work

- **Binius (2023):** Binius demonstrates the efficacy of binary tower fields for proof systems but relies heavily on degree-2 product sumchecks to handle non-linear operations. FROST-GKR diverges by directly arithmetizing high-degree operations, yielding better performance for compute-intensive primitives like Poseidon2b.

- **Lasso & Jolt (2023):** These systems use lookup arguments to handle non-linear operations (e.g., x^7 via large lookup tables). FROST-GKR avoids lookup tables entirely, utilizing the Frobenius endomorphism to prove the operation algebraically. This is particularly advantageous in $GF(2^{128})$ where lookup tables over the full field are computationally infeasible (requiring $O(2^{128})$ memory).
- **Libra / Virgo:** Early GKR implementations focused on layered circuits. FROST-GKR introduces the Shift Gadget to flatten sequential, stateful computations (like iterated hash permutations) into a unified MLE layout, a problem not directly addressed by layered GKR models.

8 Conclusion

We have presented FROST-GKR, a new construction for proving sequential hash functions over binary fields. By breaking the degree-2 barrier and directly arithmetizing the degree-7 S-box via the Frobenius endomorphism, and introducing the Shift Gadget to decouple sequential linear constraints from non-linear ones, we achieve orders-of-magnitude improvements in validity proof generation for Poseidon2b. We believe this technique is broadly applicable to other arithmetization-oriented primitives over binary tower fields.

Acknowledgements

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